

Route A: Matched Dense-Confidence Enrichment for POPE & TriviaQA — Design

MH-Neuron / Modality Taxonomy

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This note explains the “Route A” fairness fix for putting the POPE and TriviaQA enrichment legs on equal footing, per the fairness spec. It is a design description; no code has been changed and no jobs have been launched.

1 The core problem it fixes

Single-batch ablation almost never flips a *discrete* outcome (a yes/no answer, or an alias match), so the binary ΔH is $\sim$ all-zero and the top-5% “driving” set collapses to an **argsort** tie-break — the degeneracy guard fires for *both* legs. Route A replaces each leg’s discrete metric with a **continuous answer-confidence probability** read directly off the LM head, so every ablated batch moves the score a little: dense, tie-free ΔH , hence a real ranking for Fisher’s exact test.

2 What the dense metric is, concretely

When the model answers, it first assigns a **probability to each possible answer** (how confident it is), then picks the highest. For a POPE question “*Is there a cat in this image?*” with truth **no**, it might compute $P(\text{“yes”}) = 0.10$, $P(\text{“no”}) = 0.90$, and pick “no” (correct).

Binary metric (old). Look only at the final *pick*: yes or no, a 0/1 outcome. Ablate 50 neurons and re-ask: now $P(\text{“yes”}) = 0.13$, $P(\text{“no”}) = 0.87$ — still picks “no.” Binary records that the pick did not change, so effect = 0. The model leaned further toward hallucinating, but binary sees nothing. This happens for almost every batch, giving a field of zeros that cannot be ranked.

Dense metric (new). Read the *probability itself*: $P(\text{“yes”})$ went $0.10 \rightarrow 0.13$, so effect = $+0.03$ — a real, measurable number, even though the answer never flipped.

So the dense metric is simply: **measure the model’s confidence (a number in $[0, 1]$), not its yes/no verdict.**

- **POPE:** confidence = $P(\text{“yes”})$ on a no-truth question = how much it leans toward hallucinating.
- **TQA:** confidence = $P(\text{correct answer word})$ = how much it leans toward the right answer.

Average this over ~ 1000 questions, with and without the ablation; the difference is ΔH . **Binary** is $\sim$ always exactly 0 (a flat field, nothing to rank); **dense** is a small real number for every batch ($0.03, -0.01, 0.008, \dots$), so the batches that move confidence most can be ranked. The shift is from asking “did the answer change?” to “how much did the model’s confidence move?” — same ablation, more sensitive measurement.

3 Why dense fixes it, in plain terms

Think of measuring how much one lecture matters to a student.

- **Binary** asks: “did the student pass or fail?” Skipping one lecture out of hundreds almost never flips pass $\rightarrow$ fail, so for nearly every lecture the answer is “no change.” If thousands of lectures all

read “no change,” you cannot tell which one mattered — the ranking is empty, and “top 5%” just picks whichever happen to come first in the list (an artifact). That is binary ΔH : removing 50 of $\sim 11,000$ neurons rarely changes the actually-chosen word or yes/no, so almost every batch scores exactly 0.

- **Dense** asks: “did the exam score move from 91 to 89?” Even when the student still passes, you see each lecture’s effect as a small change in the score. That is the confidence probability: even when the model still says the same answer, removing 50 neurons nudges the *probability* it assigned — say from 0.91 to 0.89. The nudge is measurable and different for every batch, so the batches that move confidence most can be ranked. No mass tie, real ranking, guard off.

The fix is not a finer scalpel; it is a more sensitive ruler applied to the same cut.

4 The shared mechanism (already built for POPE)

`_AnswerLogitCapture` registers a forward hook on the model’s `lm_head`. During a single forward (the prefill, `max_new_tokens=1`) it captures the logits at the **answer position** — the last prompt token, whose next-token distribution *is* the model’s answer. This works uniformly across backends (including InternVL’s `model.chat`) because they all call `lm_head` once on the prefill. It runs *inside* the ablation context manager, so the captured logits are the **ablated** answer distribution.

5 POPE leg (exists: `halluc_measure='logit_margin'`)

Per POPE question, at the answer position:

- $P(\text{yes}) = \sigma(\text{logit}[\text{“Yes”}] - \text{logit}[\text{“No”}])$ — continuous probability the model gives the hallucinated “yes”.
- Hallucination score = mean of $P(\text{yes})$ over the **gt=no** questions (a *soft* false-positive rate).
- $\Delta H_{\text{POPE}} = \text{soft_rate}(\text{ablated}) - \text{soft_rate}(\text{baseline})$.

6 TQA leg (the new piece — in `compute_triviaqa_error_rate`)

Currently TQA generates 32 tokens and alias-matches (binary). The new dense mode mirrors POPE exactly:

- Each TQA question has gold **aliases** (e.g. `["New York", "NYC"]`); take the **first token** of each alias.
- At the answer position, $P_{\text{gold}} = \max_{\text{alias}} \text{softmax}(\text{logits})[\text{first_token}(\text{alias})]$ — the model’s confidence it is about to start the correct answer.
- Per-question **error** = $1 - P_{\text{gold}}$ (mirrors POPE’s $P(\text{yes})$, the probability of the *wrong* outcome).
- TQA score = mean error over questions; $\Delta H_{\text{TQA}} = \text{error_rate}(\text{ablated}) - \text{error_rate}(\text{baseline})$.

It reuses the *same* `_AnswerLogitCapture/lm_head` hook and `_get_tokenizer` machinery — no new generation, just one forward per (batch, question) reading the gold-token probability.

7 Why this is a *matched* pair

Both legs are now: “*answer-position softmax probability of the key answer token, aggregated over the contrastive set, differenced under ablation.*” POPE tracks $P(\text{wrong “yes”})$; TQA tracks $1 - P(\text{right answer})$. Both are continuous $[0, 1]$ confidences from the same hook, so an $\text{OR} > 1$ on one task and $\text{OR} < 1$ on the other reflects modality, not a metric-shape artifact. One minor asymmetry, documented: POPE’s $P(\text{yes})$ is a 2-way Yes/No softmax while TQA’s P_{gold} competes against the full vocabulary — but enrichment uses only the *ranking* of ΔH within a model, which is scale-invariant under Fisher, so this does not bias the comparison.

8 Sign convention (matched)

For both legs, **larger ΔH = removing this neuron damages the task more** (POPE: ablation raises $P(\text{yes})$ /hallucination; TQA: ablation raises error / lowers gold confidence). `compute_enrichment` takes the top- k *largest*, so both put the “most task-protective neurons” at the top. Identical orientation.

9 Matched intervention & enrichment path (unchanged)

- Same `ablation_worker`, same batch-of-50 gate-hook zeroing — ΔH_{POPE} and ΔH_{TQA} are computed in the *same pass* per batch, so granularity is identical.
- Both scored through the *same* `compute_enrichment` (same `top_k_pct=5`, same `flat_labels` neuron universe, same `n_random_trials=1000`, same `seed`), emitting `enrichment_results.json` + `enrichment_results_tqa.json` from one run: guaranteed parameter-matched.
- `compute_enrichment`’s statistics, top- k logic, and random baseline are **not** touched (spec hard constraint).

10 Resolution: 50 neurons at a time

Both legs share the same intervention: each ablation zeros a contiguous block of 50 neurons at the gate hook. Per block, in one pass,

$$\Delta H_{\text{POPE}} = \overline{P(\text{yes})}_{\text{gt=no}}^{\text{ablated}} - \overline{P(\text{yes})}_{\text{gt=no}}^{\text{baseline}}, \quad \Delta H_{\text{TQA}} = \overline{(1 - P_{\text{gold}})}^{\text{ablated}} - \overline{(1 - P_{\text{gold}})}^{\text{baseline}},$$

i.e. ΔH_{POPE} is the change in mean $P(\text{yes})$ over the `gt=no` questions, and ΔH_{TQA} is the change in mean $(1 - P_{\text{gold}})$ over the TQA questions. The block’s ΔH is then split equally among its 50 neurons, so all 50 share one score (but keep their own modality labels). The effective resolution is therefore **per-50-neuron block**, not single-neuron — and the dense metric removes the degeneracy at exactly that block level: binary makes nearly every block read 0 (unrankable); dense gives every block a distinct continuous value (rankable). The only leftover tie is the single block straddling the top-5% cutoff (~ 50 neurons), far below the guard’s $>50\%$ -tied threshold. Single-neuron ablation would be $50\times$ more forward passes *and* would unmatch the two legs, so it is intentionally not used.

11 Contrastive filtering parity (spec §2)

Verify `build_contrastive_triviaqa_set` uses the same K -samples / temperature / consistency criterion / cap as POPE’s $K = 10$ at $T = 0.7$, ~ 1000 cap — align or document any difference, and report both retained-set sizes side by side.

12 What actually runs on waic-risk

One control flag (`-halluc_measure logit_margin`) flips *both* legs to dense. Then a fresh step-10 per model: `-long-desc` ($\alpha=0.001/\text{max-300 labels}$) `-halluc_triviaqa -contrastive`, per-layer ablation jobs (36 for qwen-3b) recomputing dense ΔH for both legs $\rightarrow$ aggregate $\rightarrow$ matched enrichment.

- **qwen2.5-vl-3b: clean** — no active step-10 jobs; runs immediately.
- **internvl & llava-ov: collision** — their binary campaign is live and writes the same `ablation_scores*.json` paths. The dense runs are routed to a **separate (suffixed) output location** so the running campaign is neither clobbered nor raced; existing binary results stay intact.

13 Code changes (scoped)

1. `compute_triviaqa_error_rate`: add the dense gold-confidence mode (the only substantive new code).
2. Thread the measure into the TQA call in `ablation_worker` (POPE side already wired).

3. A small `run_pipeline.sh` plumbing line so `HALLUC_MEASURE=logit_margin` reaches both legs.
4. Verify/align the TQA contrastive builder parameters.

No changes to enrichment statistics, labels, or descriptions.

14 Risks

1. It is a real GPU campaign (3 models $\times$ per-layer) on a busy cluster.
2. `internvl/llava-ov` need the separate-output handling to avoid the live campaign.
3. The dense TQA metric is a *new* signal — validate it is non-degenerate (guard off) on qwen-3b first before trusting/expanding.
4. Results may still show no clean dissociation; per the spec that is a valid outcome, reported as-is. This is a neutral fairness fix, not an attempt to produce a dissociation.